

COURSE 3 - DATA ENGINEERING 2026**C3-083 - StoreLake Fabric Mini-Lab**

Independent integrated Fabric build using fresh StoreLake data. Limited hints only after a genuine blocked attempt.

Artifact	C3-083
Status	2026 - LEARNER EDITION
Phase	C3-07 Cloud & Microsoft Fabric Data Engineering 15 lessons 50-70 focused hours
Prerequisite	C3-06 Gate PASS; C2B cloud foundations are recalled, while Fabric implementation is taught from first principles.
Current lane	Microsoft Fabric docs refreshed 2026-09-11 trial 60 days F4/F64 up to 1 TB OneLake
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided Fabric action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> security/production reasoning -> evidence. A portal screenshot is not mastery unless the underlying data, identity, lifecycle and recovery claim is independently verified.

Scenario - StoreLake

StoreLake is moving local order files into Fabric. Build a governed solution for full + incremental orders, trusted Lakehouse tables and a SQL serving layer, then prove lifecycle, security, monitoring and capacity-aware operation.

Required build

- Create/plan workspace and Lakehouse with Bronze/Silver contracts.
- Use Data Factory movement/orchestration for initial landing and notebook/Spark transform. Consider Copy Job where it is the simpler fit; explain your final choice.
- Apply incremental file with stable key + updated_at state/MERGE behavior and prove rerun idempotence.
- Create Warehouse or Lakehouse SQL serving object for daily store revenue and reconcile it.
- Put supported definitions/schema under branch/PR/Dev-Test lifecycle design; remap environment-specific references.
- Define analyst access with control-plane + OneLake/SQL data-plane controls and prove one blocked consumer test.
- Save run/query/reconciliation/monitoring evidence and diagnose one injected failure.
- Create a capacity-aware schedule and map evidence to the current DP-700 domains.

Completion contract

PASS requires every required competency below to PASS and zero unresolved critical failures. A local/design-only result remains Developing for any competency that requires Fabric execution.

Required competencies

- Architecture and platform choice
- Ingestion + lakehouse + incremental correctness
- Serving/query correctness
- Lifecycle/change safety
- Security/governance consumer test
- Observability/recovery
- Capacity/operability
- Reproducibility/handoff

Critical failures

- Fabricated cloud execution/monitoring/security evidence
- Secret or credential exposed in source/evidence
- Broad privilege used to bypass a least-privilege requirement
- Incremental/rerun logic duplicates or regresses trusted state
- Security claim without testing the intended consumer identity
- Claiming Git/deployment is a backup of OneLake tables/files
- Failed run/query treated as success or unsafe recovery corrupts data
- Protected review opened before genuine attempt

Evidence bundle

Architecture + pipeline/job/notebook/query definitions + source/target controls + incremental rerun proof + security allow/block test + monitoring/recovery + release plan + capacity note + README/handoff. Real tenant evidence is mandatory for live-platform competencies.